WEAKLY ETALE MORPHISMS ARE FORMALLY UNRAMIFIED: THE DIFFERENTIALS VANISH

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19041-19045 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19047-19054, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Fifth member of the weakly etale series (L16089-L16091, L16119); first consequence-direction card of that toolkit.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma formally-unramified-differentials; chapter lemma weakly-etale-characterize (card L16119); Morphisms, lemma differentials-affine; More on Algebra, lemma formally-unramified (ring case).

Source reference. Original lemma 02H9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 898–901. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 094S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18935–18950. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 092M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19054–19054. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19054–19054. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a weakly étale morphism of schemes. Then f is formally unramified, i.e., $\Omega_{X/Y} = 0$.*

Proof. Recall that f is formally unramified if and only if $\Omega_{X/Y} = 0$ by Lemma 02H9. Via Lemma 094S and Morphisms, Lemma 01UT this follows from the case of rings which is More on Algebra, Lemma 092M. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.